

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS802A - ENERGY EFFICIENCY ACOUSTICS AND DAYLIGHTING IN BUILDING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 In tropical environments, buildings are typically designed to:
 - A) Retain internal heat
 - B) Maximize solar gain
 - C) Encourage cross ventilation
 - D) Minimize roof overhangs
- 2 Which of the following factors most directly affects thermal comfort in a built environment?
 - A) Paint color
 - B) Air temperature
 - C) Wall thickness
 - D) Room volume
- 3 The main mode of heat gain in a building due to sunlight is:
 - A) Conduction
 - B) Convection
 - C) Radiation
 - D) Evaporation
- 4 Steady-state heat transfer assumes that:
 - A) Heat input changes constantly
 - B) Temperature changes with time
 - C) Temperature remains constant over time
 - D) Heat flow is random
- 5 The transmission matrix method is primarily used to:
 - A) Predict lighting
 - B) Model acoustic behavior
 - C) Compute heat flow through building elements
 - D) Calculate ventilation

- 6 Energy-efficient buildings are often oriented to:
- Maximize north-facing walls
 - Minimize east-west exposure
 - Eliminate natural light
 - Increase glass area on all sides
- 7 The primary purpose of natural ventilation in buildings is to:
- Block sunlight
 - Reduce noise
 - Improve indoor air quality
 - Enhance structural stability
- 8 Protection against external noise is best achieved by:
- Increasing window size
 - Using lightweight materials
 - Providing insulation and buffer zones
 - Removing vegetation
- 9 In daylight design, fenestration refers to:
- Wall color
 - Type of ventilation
 - Arrangement and size of window openings
 - Roof structure design
- 10 Which among the following is the unit of Illumination?
- Lux
 - Candela
 - Lumen
 - Candela/m
- 11 Airborne noise is best controlled by:
- Sound barriers and dense materials
 - Increasing room temperature
 - Enhancing lighting
 - Removing windows
- 12 Daylight factor is defined as the ratio of:
- External to internal heat
 - Internal to external light level
 - Window to wall area
 - Artificial to natural light
- Q2. Solve the following.
- Discuss in brief- Impact of environment on building. 6
 - Explain Tropical environments and site environments briefly. 6
- Q3. Solve the following.
- Enlist and describe factors affecting on human comfort. 6
 - Explain in detail human response to thermal environment. 6
- Q4. Solve Any Two of the following.
- Illustrate Heat Flow in buildings. 6
 - Describe steady and periodic heat transfer in buildings. 6
 - Explain the Admittance Method in the context of acoustics analysis. 6

Q5. Solve Any Two of the following.

- A) Describe Structural control and design for energy efficiency. 6
- B) Explain Mechanisms of Natural ventilation. 6
- C) How to increase natural ventilation in a building? 6

Q6. Solve Any Two of the following.

- A) What are the protective measures against external noise? 6
- B) Write a short note on Basic acoustics and noise. 6
- C) Discuss the role of daylight prediction models, considering the unique characteristics of the Indian sky, in the design of fenestration for buildings in India. 6

*** End ***